

Topic 2: ESP32

The Cost-Effective Internet of Things (IoT) Platform

Questions

- What makes a product successful?
 - What makes Japanese cars successful in the US market?
 - Then, what makes ESP32 successful?

My answer is: **Cost-Effective.**

- It's not about the best quality or the most features, but about the best value for the price.
- Even though Japanese cars may not be the best in terms of performance, they offer a good balance of quality and affordability.
- Similarly, ESP32 may not be the most powerful or feature-rich microcontroller, but it is good enough for most IoT applications.

What is the problem that ESP32 solves?

- Before ESP32, we had to choose between expensive microcontrollers with WiFi capabilities or cheap ones without WiFi.
- ESP32 provides a cost-effective solution by integrating WiFi and Bluetooth capabilities into a single chip, making it accessible to a wide range of developers and hobbyists.

How does ESP32 achieve cost-effectiveness?

- Mass production and widespread adoption have driven down the cost of ESP32 chips.
- The success of its predecessor, ESP8266, paved the way for ESP32's popularity and affordability.
- Then, ESP32 improves by offering more features and better performance while maintaining a low price point.

The Software Ecosystem

- ESP32 supports ESP-IDF, the official development framework, which provides a comprehensive set of tools and libraries for programming ESP32.
- However, many developers prefer using the Arduino IDE for ESP32 programming due to its simplicity and large community support.

The ESP32 Variants

- There are many variants of ESP32 boards available in the market, each with different features and specifications.
- For example, ESP32C6 is a low-cost and focus on Home automation with Zigbee support, while ESP32-S3 is a high-performance variant with AI capabilities.

Choosing the **Right** ESP32 Variant

The ESP32 family: One size doesn't fit all.

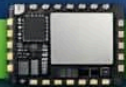


ESP32-C3

Low-Power IoT

Wi-Fi + BLE | RISC-V

Smart plugs & sensors.



ESP32-C6

Smart Home Ready

Wi-Fi 6 + 802.15.4

Matter & Thread.



ESP32-H2

Mesh Networks

BLE + 802.15.4

Zigbee & Thread.

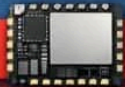


ESP32-S3

AI & USB

Wi-Fi + BLE | Dual-Core

Displays & Voice.

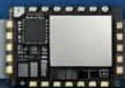


ESP32-C5

High Performance Wi-Fi

Dual-Band Wi-Fi 6

Industrial IoT.



ESP32-P4

No Wireless

High-Perf RISC-V

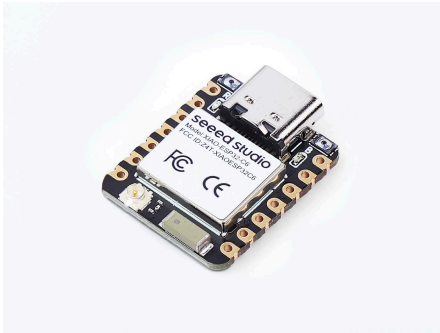
LCD & Control Systems.



There's no **"best"** ESP32, only the **right ESP32** for the job.

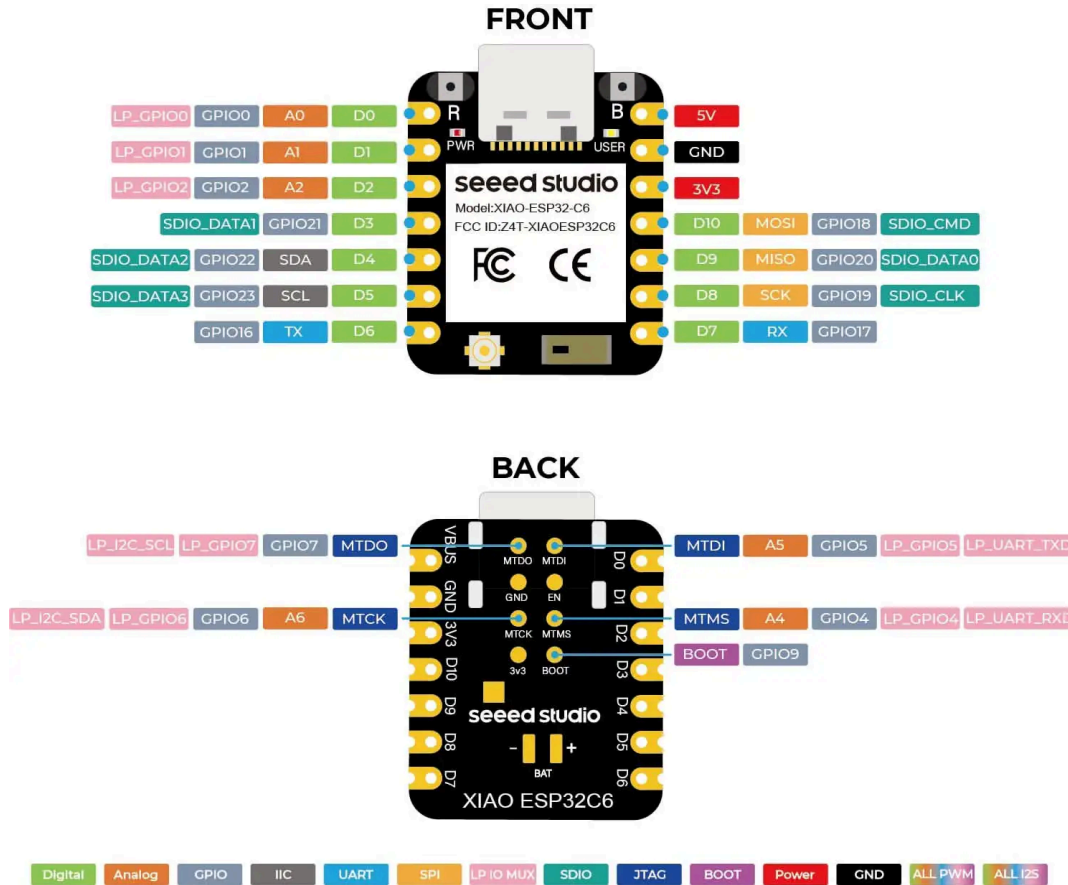
💬 Comment your favorite MCU and why you prefer using it!

Using ESP32C6



We are using Seeed Studio XIAO ESP32C6 for our project, which is a compact and cost-effective board with Zigbee support, making it ideal for home automation applications.

XIAO ESP32C6 has limited number of external I/Os, so pins have duplicate functions.



Using Arudion IDE for ESP32 Programming

- We need to teach Arudion IDE the JSON information about the board

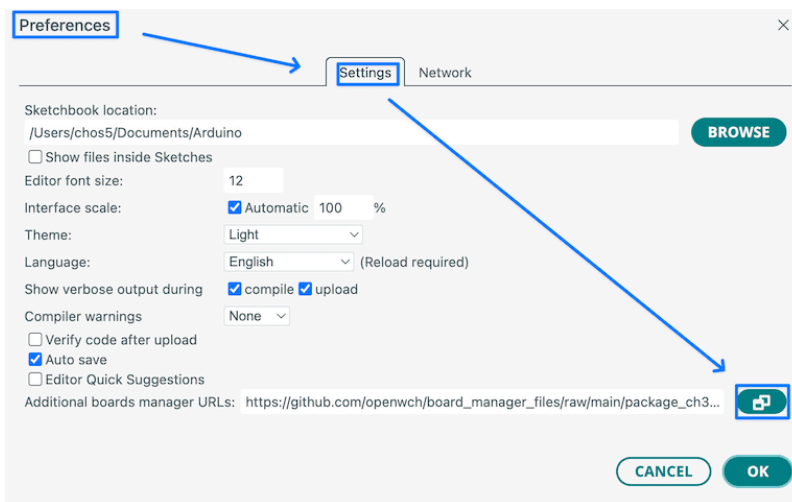
https://espressif.github.io/arduino-esp32/package_esp32_index.json

Use this page to learn how to add ESP32 support to Arduino IDE:

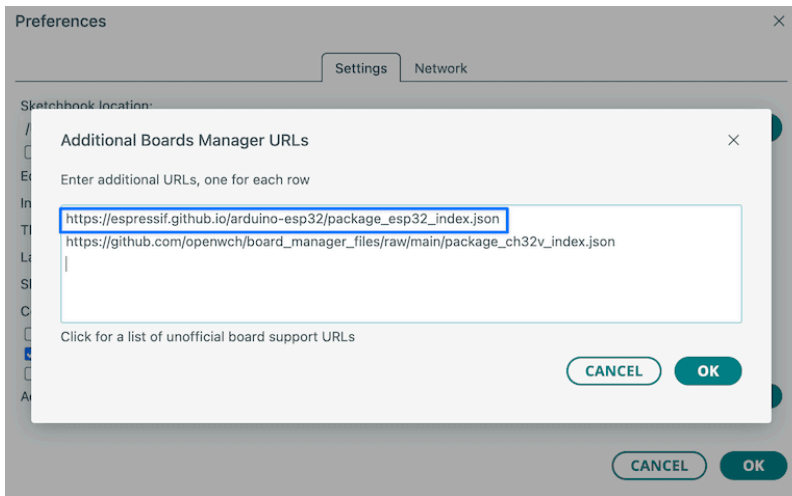
https://wiki.seeedstudio.com/xiao_esp32c6_getting_started/

Board Manager

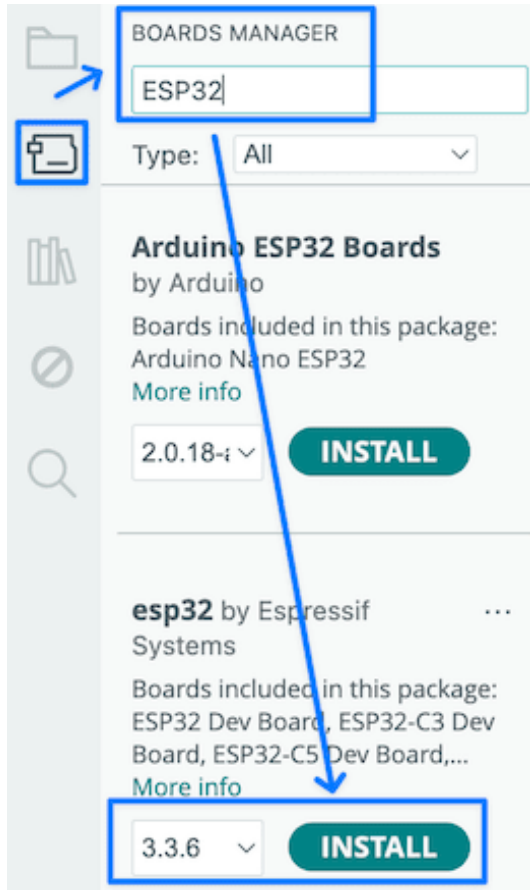
Preference -> Settings -> Board Manager



Add https://espressif.github.io/arduino-esp32/package_esp32_index.json to the board manager.

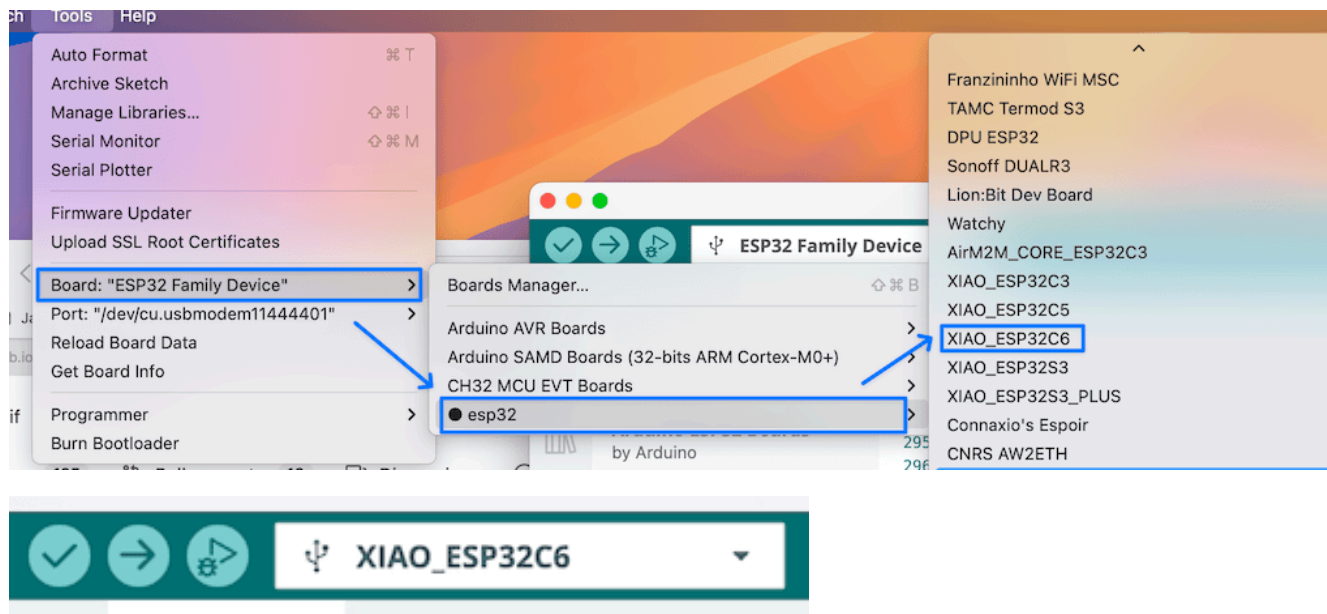


Open Board Manager and install "esp32 by Espressif".

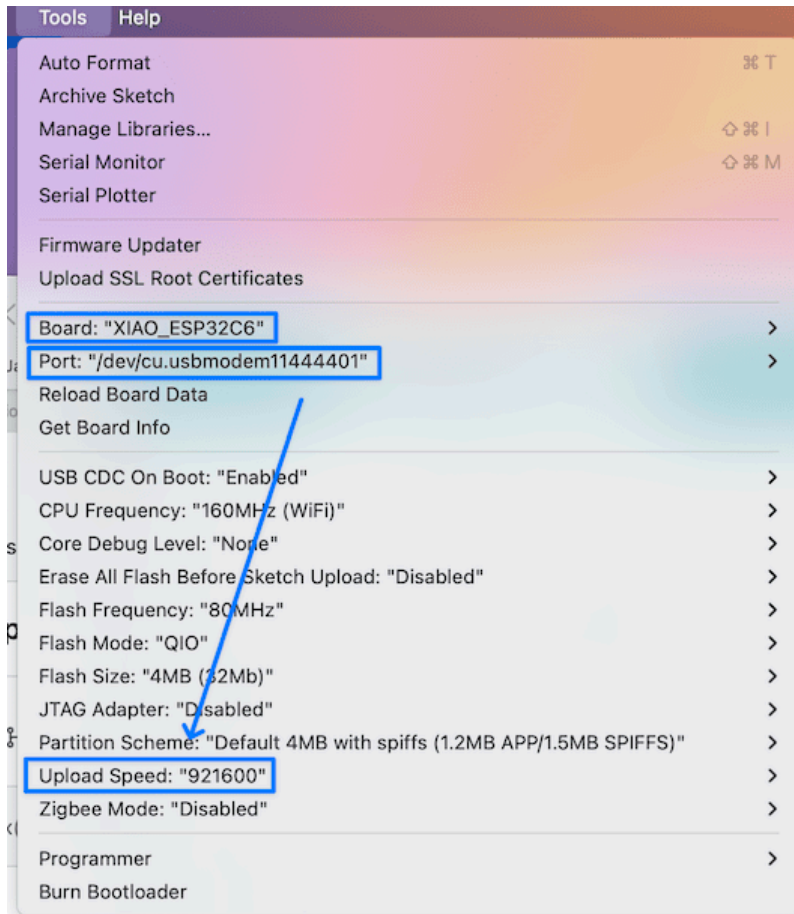


[https://github.com/espressif/arduino-esp32?
tab=readme-ov-file#supported-chips](https://github.com/espressif/arduino-esp32?tab=readme-ov-file#supported-chips)

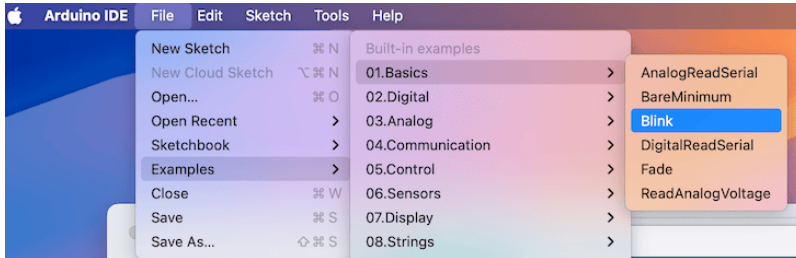
From the Board menu, select esp32 -> XIAO ESP32C6.



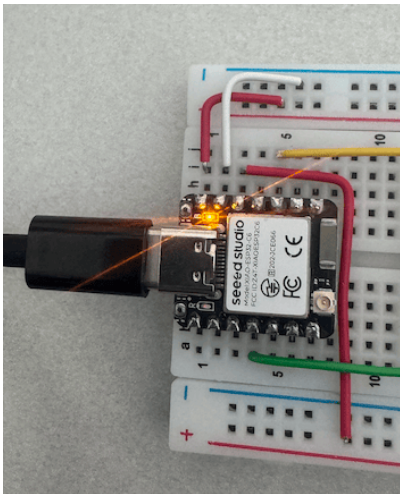
Check the board, port, and baud rate.



Run the Blink Example



Make sure the Arduino IDE compiles, links, and uploads the binary (HEX) to the ESP32 board.



Test Other Examples

There are many ESP32C6 examples that you can use:

1. ESP32C6 as the web server.
 - Any IoT device can access ESP32C6 to get information.

2. ESP32C6 as the web client that accesses the web server.

- We need a web server that ESP32C6 should send the information to.
- ESP32C6 accesses the web server to send the information.

Choose the example for your project start testing it.